Approximating the Score Function using Semi-Discrete Optimal Transport

Curves and Surfaces 2026

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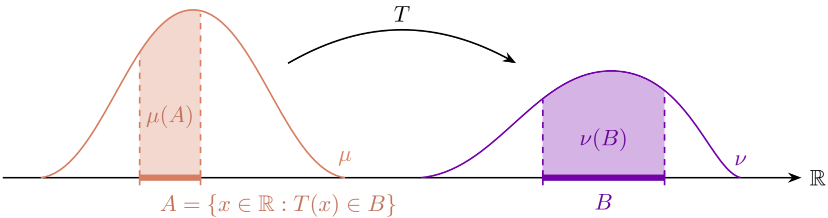
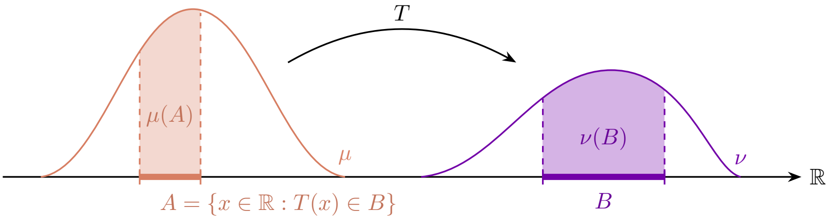
Quick reminders on Optimal Transport 1/2



- Given ground cost $c(x, y) = \|x - y\|^2$
- Among all maps T satisfying $\forall B,\ \nu(B) = \mu(T^{-1}(B))$
- Find the one minimising the average transport cost

$$W_2(\mu, \nu)^2 := \inf_{T\,:\,T_{\#}\mu=\nu} \int_{\mathbb{R}^d} \|x - T(x)\|^2 \mu(\mathrm{d}x)$$

Quick reminders on Optimal Transport 2/2



- Kantorovich duality

$$\inf_{T\,:\,T_{\#}\mu=\nu} \int_{\mathbb{R}^d} \|x-y\|^2 \mu(\mathrm{d}x) = \sup_{\phi} \int \phi(x)\mu(\mathrm{d}x) + \int \phi^c(y)\nu(\mathrm{d}y),$$

where $\phi^c(y) := \inf_x c(x,y) - \phi(x)$.

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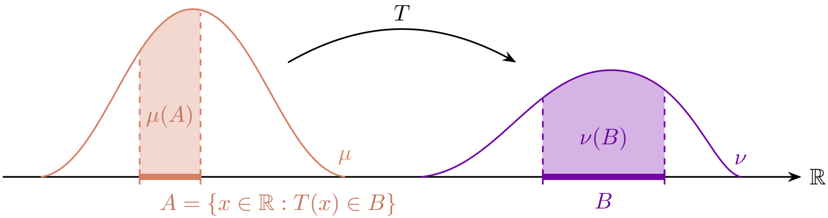

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- Let $c(x,y) = \alpha\|x-y\|^2$

$$\nabla_x c(x, T^\star(x)) = \nabla\phi^\star(x) \iff T^\star(x) - x = \boxed{y - x = -\frac{1}{2\alpha}\nabla\phi^\star(x)}$$

Displacement

What is the score function?

- **Forward process:** Ornstein–Uhlenbeck (OU) process:

$$\mathrm{d}X_t = -X_t \mathrm{d}t + \sqrt{2}\mathrm{d}\omega_t, \quad X_0 \sim \mu, \quad t \in [0, T].$$

Let $\rho_t := \mathrm{Law}(X_t)$. Then ρ_t solves the Fokker–Planck equation

$$\partial_t \rho_t(x) = \Delta \rho_t(x) + \mathrm{div}(\rho_t(x)x). \tag{$\star$}$$

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- **Macroscopic POV:** Equation $(\star)$ is the 2-Wasserstein Gradient Flow of the relative entropy $\mathcal{H}(\rho|\gamma) := \int \frac{\|x\|^2}{2}\rho + \int \rho \ln(\rho)$. It means $(\star)$ can be written as

$$\partial_t \rho_t(x) = -\nabla \cdot (\rho_t(x) v_t(x)),$$

  where $v_t(x) = -x - \nabla \ln(\rho_t(x))$.

Using OT to learn the score 1/2

- Given a time-step $\tau > 0$, such a gradient flow can be discretised in time using the **JKO scheme** (or Implicit Euler scheme, or Proximal step)

$$\begin{cases} \rho_0 = \mu \\ \rho_{n+1}^\tau := \arg\min \mathcal{H}(\rho|\gamma) + \dfrac{1}{2\tau} W_2(\rho_n^\tau, \rho)^2 \end{cases}$$

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- **At step n:** We show that

$$\rho_{n+1}^\tau \propto \exp\left(-(\phi^\star)^{c/2\tau}\right) \mathrm{d}\gamma$$

where $\phi^\star$ is a Kantorovich potential satisfying

$$\phi^\star \in \arg\min_\phi \tilde{J}(\phi) := \ln\left(\int e^{-\phi^{c/2\tau}} \mathrm{d}\gamma\right) - \int \phi \mathrm{d}\rho_n^\tau.$$

Using OT to learn the score 2/2

- The optimal potential $\phi^\star$ satisfies ($\alpha = \frac{1}{2\tau}$ here)

$$\underbrace{\boxed{-\tau\nabla\phi^\star(x) = y - x}}_{\textbf{Displacement}}$$

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$$-\tau\nabla\phi^\star(x) = \tau v_t(x) = -\tau(x + \nabla \ln\left(\rho_{n+1}^\tau(x)\right)) \iff \boxed{\nabla \ln\left(\rho_{n+1}^\tau(x)\right) = \nabla\phi^\star(x) - x}.$$

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- We'll approximate the score by approximating the Kantorovich potential $\phi^\star$, that minimises the functional:

$$\tilde{J}(\phi) = \ln\left(\int e^{-\phi^{c/2\tau}}\right)\mathrm{d}\gamma - \int \phi \mathrm{d}\rho^\tau_n$$

General contributions

Proposition *(G.)*

The functional $\tilde{J}(\phi) := \ln\left(\int e^{-\phi^{c/2\tau}} \mathrm{d}\gamma\right) - \int \phi \mathrm{d}\rho_n^\tau$ is convex.

General contributions

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Theorem *(G.)*

Let $J(\phi) := \int e^{-\phi^{c/2\tau}}\mathrm{d}\gamma - \int \phi \mathrm{d}\rho_n^\tau$. Then,

$$\arg\min_{\phi} J(\phi) \subseteq \arg\min_{\phi} \tilde{J}(\phi) \quad \text{and} \quad \arg\min_{\int e^{-\phi^{c/2\tau}}\mathrm{d}\gamma=1} \tilde{J}(\phi) = \arg\min_{\phi} J(\phi).$$

Pro: J is "SGDisable"

Con: one more parameter to estimate (the normalisation constant!)

Stochastic gradient descent 1/3

- $J(\phi) = \mathbb{E}_{y \sim \gamma}\left[\exp\left(-\phi^{c/2\tau}(y)\right) - \int \phi(x)\rho_n^\tau(\mathrm{d}x)\right]$

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- Choice of parametrisation: semi-discrete optimal transport using

$$\hat{\rho}_n^\tau = \frac{1}{N}\sum_{i=1}^{N} \delta_{\hat{x}_i^{(n)}} \rightsquigarrow \begin{cases} \phi(x;\theta) = \theta^\top \Phi(x) \\ \Phi(x) = (\Phi_1(x), \dots, \Phi_N(x)) \\ \Phi_i(x) = \mathbb{1}_{\{x = \hat{x}_i^{(n)}\}} \end{cases}$$

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- $G(\theta) := J(\phi(\cdot;\theta)) = \mathbb{E}_{y\sim\gamma}\left[\exp\left(-\min_{1\le i\le N}\left\{\frac{c(\hat{x}_i^{(n)}, y)}{2\tau} - \theta_i\right\}\right) - \frac{1}{N}\sum_{i=1}^N \theta_i\right]$

Stochastic gradient descent 2/3

Projected SGD algorithm $\left(c_\tau(x,y) := \frac{c(x,y)}{2\tau}\right)$ for some R_{proj}:

$$\begin{cases} \text{pick } y \sim \gamma \\ \text{let } i := \arg\min_j \ c_\tau(\hat{x}_j^{(n)}, y) - \theta_j \\ \text{define } e_i \text{ as the } i\text{-th canonical basis vector of } \mathbb{R}^N \\ \theta^{(k+1)} := \text{proj}_{B(0,R_{\text{proj}})}\left(\theta^{(k)} - \alpha_k \underbrace{e^{\theta_i^{(k)} - c_\tau(\hat{x}_j^{(n)}, y)}}_{\text{A priori } \textit{unbounded}!} e_i + \frac{\alpha_k}{N}\mathbb{1}\right) \end{cases}$$

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- Gradient bounded by $e^{R_{\text{proj}}}$

Stochastic gradient descent 3/3

Theorem *(G.)*

- $\alpha_k = \frac{C}{\sqrt{k}}, \quad \hat{\theta}_K := \frac{1}{K} \sum_{k=0}^{K} \theta^{(k)}$

Then, it holds that

$$0 \leq \mathbb{E}[G(\hat{\theta}_K)] - G(\theta^\star) \leq \frac{2R_{\text{proj}}^2}{CK} + \frac{B^2 C}{\sqrt{K}} = \mathcal{O}(K^{-\frac{1}{2}}),$$

where $\theta^\star$ denotes the unique minimiser of G.

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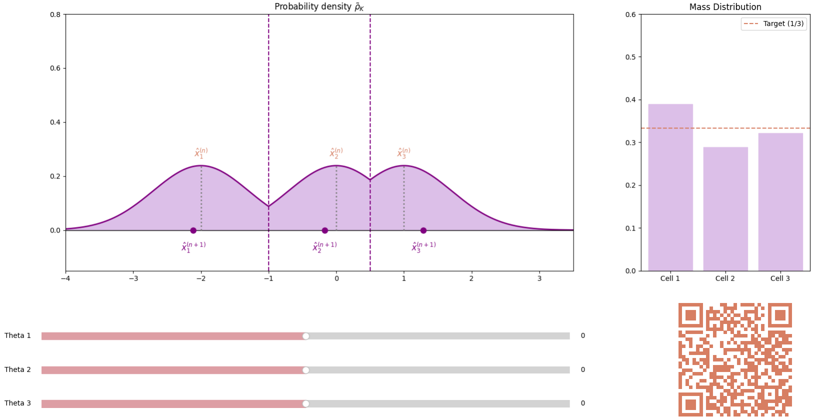

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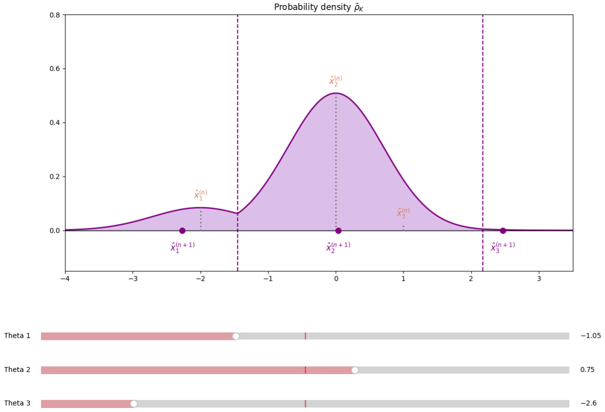

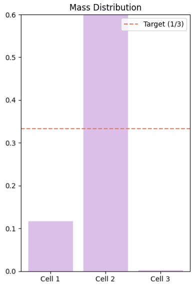

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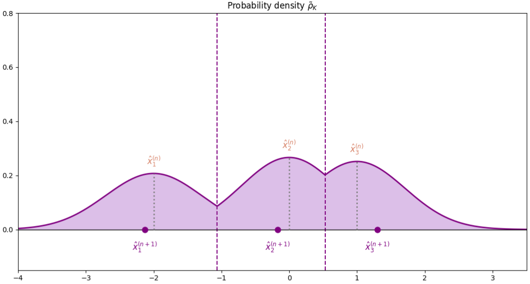

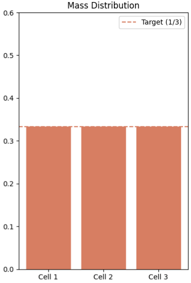

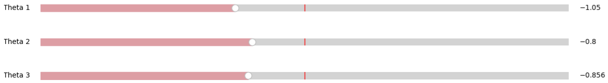

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Iterating the whole procedure

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\hat{\rho}_0 \arrow[r, "\mathrm{JKO}_\tau"] \arrow[dr, "\mathcal{T}"'] & \rho_1^\tau \arrow[r, "\mathrm{JKO}_\tau"] & \rho_2^\tau \arrow[r, "\mathrm{JKO}_\tau"] & \cdots \arrow[r, "\mathrm{JKO}_\tau"] & \rho_n^\tau \\
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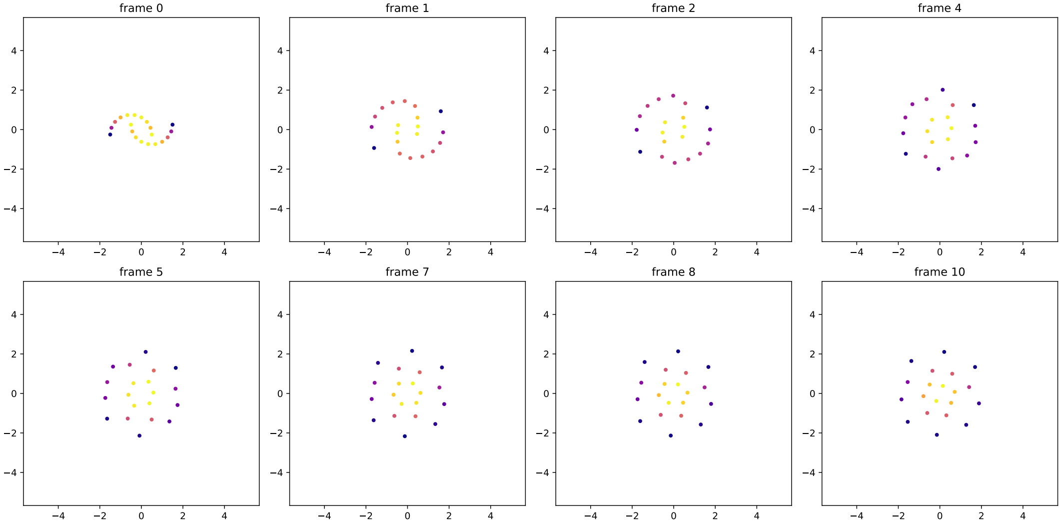
Some "animations"



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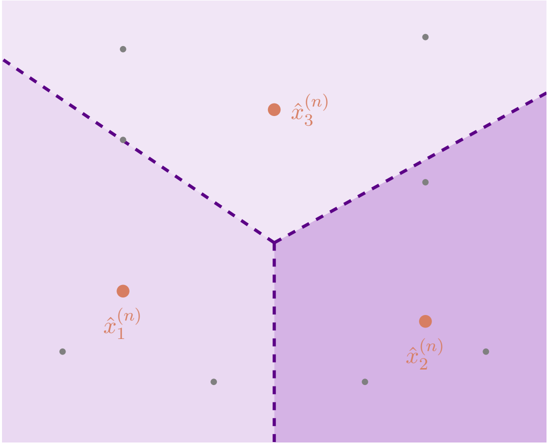

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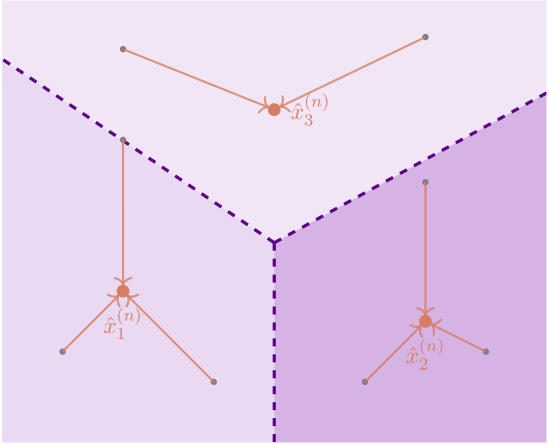

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Thank you for your attention!